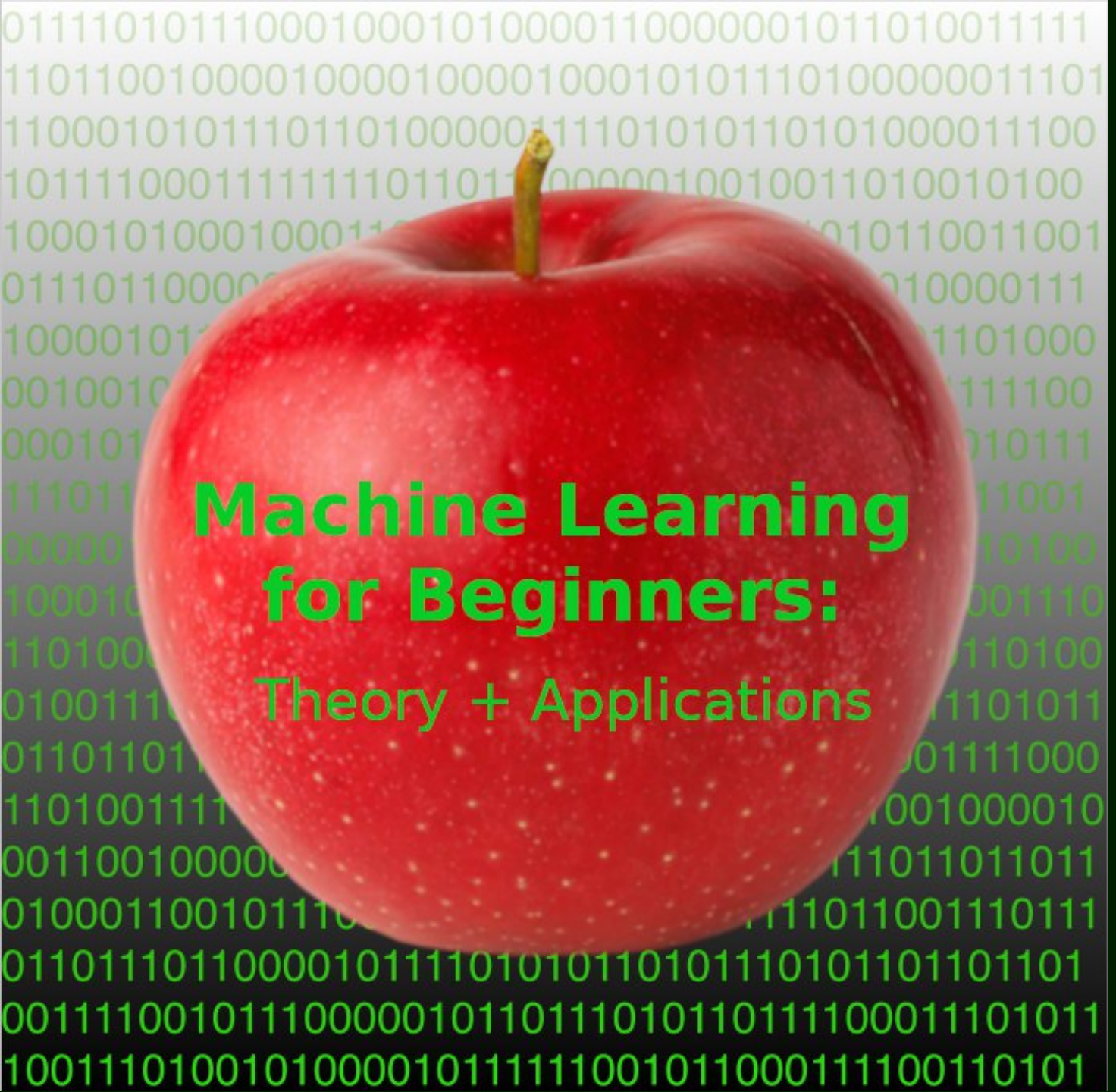


University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

Title:

"Machine Learning for Beginners: Theory and Applications"

(Available online in Canvas)

Lecturer:

Dr. Rodrigo Q. Albuquerque

First Lecture

Registration

- **Don't have a CANVAS acount yet?**
  Sign up at https://k12.instructure.com/register using the following join code: T68FJP
  IMPORTANT: When inside CANVAS, go to Account => Settings and add (on the top/right) your own email.

- **Already have a CANVAS account?**
  https://k12.instructure.com/enroll/T68FJP
  (Save the bookmark for later access!)

(Mobile registration)

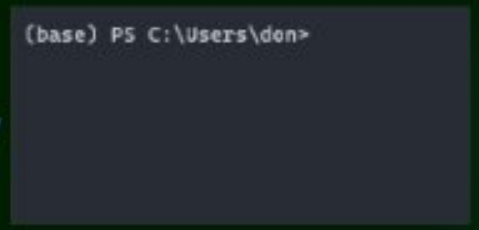
Be sure to have jupyter notebook locally installed
OR
a gmail account to use google colab online

Local Python Installation

1) Download and execute the installation file, then follow the instructions there.

https://docs.conda.io/en/latest/miniconda.html Click on Download (upper/right corner), then on Download Miniconda installer at the bottom of the page.

2) Windows: Launch the *Anaconda Powershell Prompt*
Mac & Linux: Open a terminal



This is a black/gray screen containing "(base) ...."

3) Install Jupyter notebook and some needed libraries:
conda install jupyter numpy matplotlib pandas

4) Launch the jupyter notebook (having google chrome is preferred over firefox):
jupyter-notebook

5) After using Jupyter Notebook, close its window (previously open in Google Chrome or Firefox), then click on the black Anaconda Powershell Prompt terminal and press CONTROL + C to close the prompt (you should see "(base) ...." again)

Topics

Week 1: Introduction
ML types, data, ethics, case studies

Week 2: Preprocessing & *k*NN
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross validation

Week 3: Linear Regression
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

Week 4: Classification
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

Week 5: Random forests
Decision trees, random forest, classification and regression applications

Week 6: PCA / KMeans
Dimensionality reduction, clustering

Week 7: Extra exercises

Week 8: Simple ANNs
Feedforward networks, activation functions, basic training

Week 9: Gaussian Processes
Kernels, gaussian processes

Week 10: Design of Experiments
DoE, random design, active learning, bayesian optimization

Weeks 11-12: Student projects

Bibliography

[1]* Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow (2019) O'Reilly Media, Inc.

[2] C. M. Bishop, Pattern Recognition and Machine Learning (2006) Springer.

[3] M. P. Deisenroth, Mathematics for Machine Learning (2020) Cambridge University Press (free PDF: https://mml-book.github.io/book/mml-book.pdf)

[4] Bill Lubanovic, Introducing Python (2015), O'Reilly Media, Inc.

[5] Paul Deitel, Harvey Deitel, Python for Programmers (2019) Pearson Education, Inc.

[6] Datasets and competitions for ML: www.kaggle.com

[7] ML tutorial: https://machinelearningmastery.com/machine-learning-in-python-step-by-step/

Exam

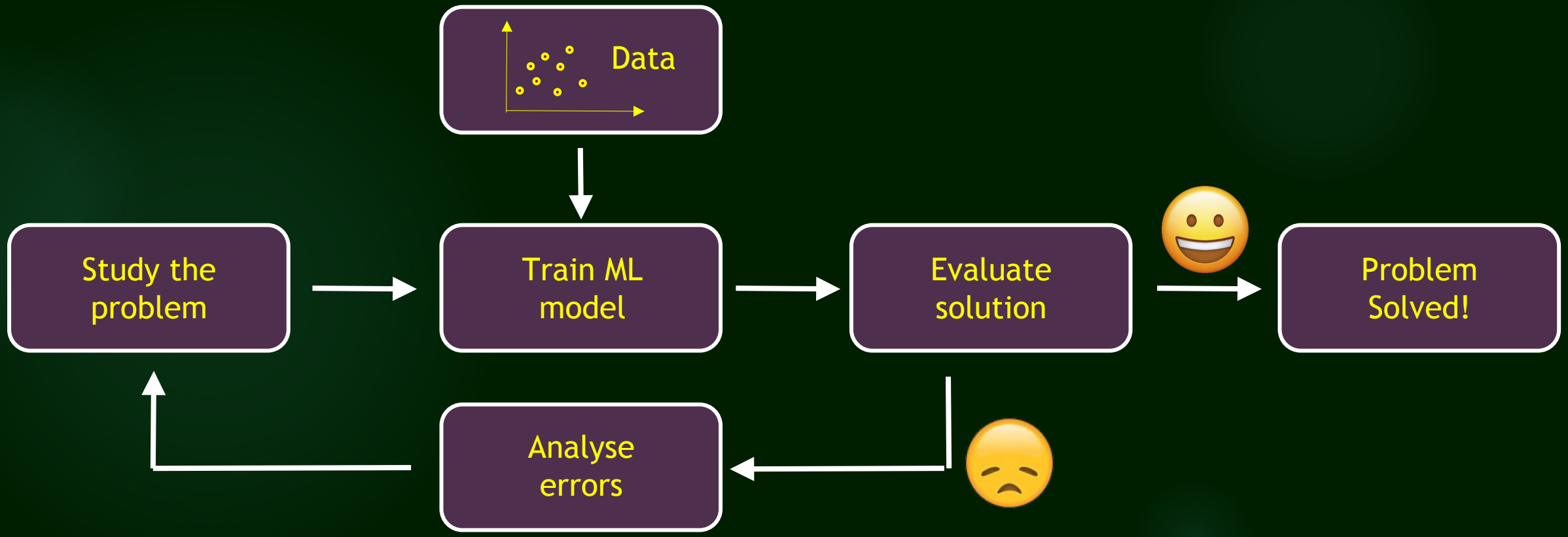
- Date and times: see Section “Lectures Overview” in the online Canvas course
- Choose any dataset (e.g., check kaggle.com) and screen at least these five regression models: RF, SVR, KNN, LASSO, GP. Present the results in a seminar of about 15 min (+/- 2min) using PowerPoint slides (what to show => see “Lectures Overview” in Canvas)
- To participate in the exam and get ECTS credits, the participants must enroll officially in the exam in CMLife.
- Participants not interested in ECTS credits can get a certificate of participation in the ML course, provided they also take part in the final exam and are approved.

Introduction

Machine Learning (ML)

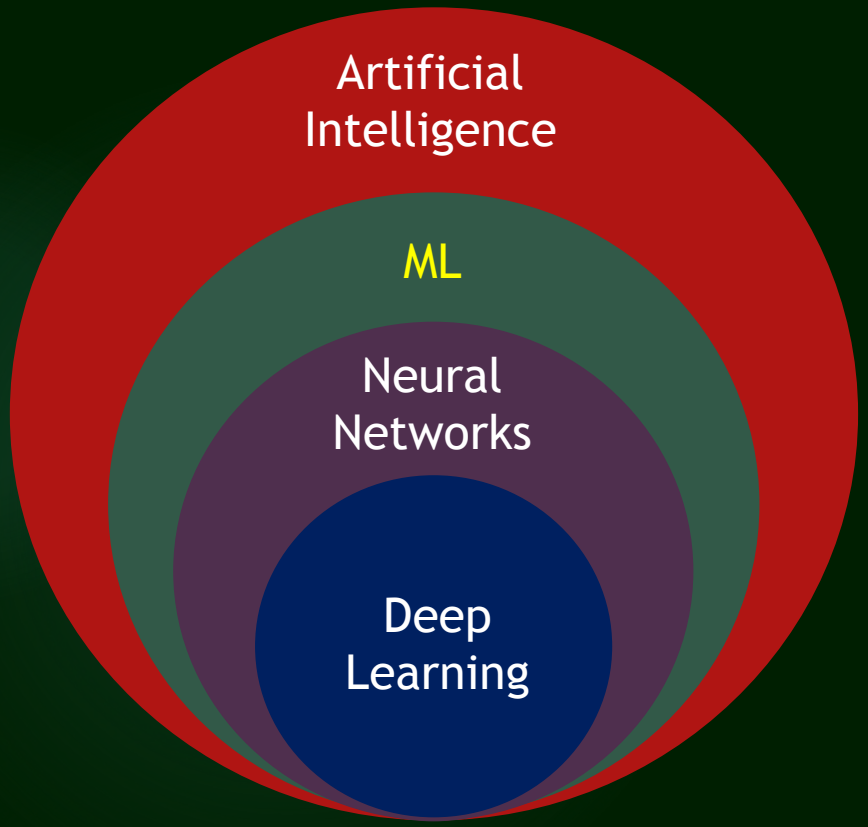
”Field of study that gives computers the ability to learn without being explicitly programmed."

(Arthur Samuel, 1959)



Introduction

General facts…



According to a recent study,[1]

- About 73 % of big German companies (>10k employees) and ca 59 % of small ones use ML daily
- Companies get return of ML use already after only 3 months!
- The market is missing ML specialists!

[1] "Studie Machine Learning 2021" – IDG research Services

Introduction

Introduction

Python

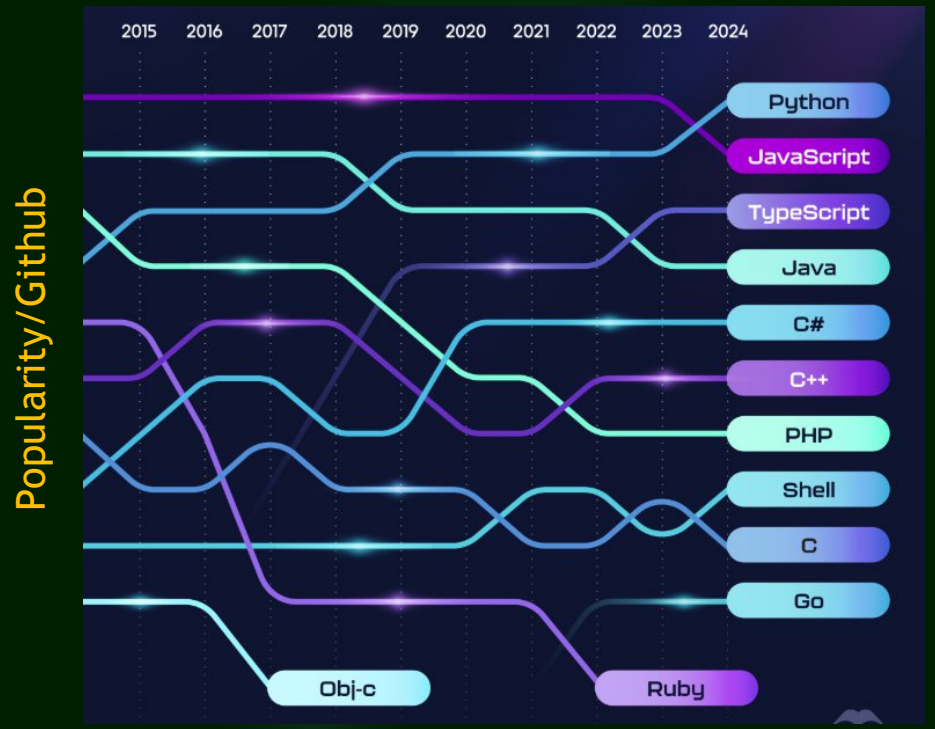
Easy, versatile and powerful object-oriented programming language.

Applications:

AI, ML, data science, web, database, simulations, etc.

Advantages of Python:

- It runs in Linux, Mac, Windows, Unix (already installed)
- It is free and open-source
- It has a GIANT community helping and further developing
- It has a large (and growing) number of very useful libraries
- It has excellent readability and is well-structured



Visit our Python course at UBT!

"Python and data tools for non-programmers"

ML Types

Supervised Learning

Regression
Classification

Material characterization
(Type prediction)

Material property
(Density prediction)

Unsupervised Learning

Clustering
Dim. Reduction

Product customer segmentation

Preprocessing for supervised learning

Semi-supervised Learning

Clustering
Classification

Speech analysis

Webpage classification

Reinforcement Learning

Control
Classification

Self-driving cars

Optimization

Supervised ML

Target properties are known. The model is trained to predict such properties.

Properties can be:

1) Continuous. Our model is trained to perform regression

E.g.: 5.4721 or 6.1304 (= Density of my material)

2) Discrete. Our model is trained to perform classification

E.g.: ”Bad” or ”Good” (= Quality of my material)

Data

Data is essentially information (anything that can be recorded):

- Numbers, text, images, sounds, etc. (sources: web, databases, sensors, equipment, etc)

Structured data

Age	Education time	Salary
20	2	17000
45	10	37000
16	0	12000
34	5	26000
27	3	22000

Examples: Excel files, csv files, databases, etc.

Python script

Unstructured data

"We examined data from five individuals, looking at their age, years of education, and reported salaries. A 20-year-old with two years of education earned $17,000, while a 45-year-old with ten years of education reported a salary of $37,000. A 16-year-old with no formal education earned $12,000. Another individual, aged 34 with five years of education, earned $26,000. Finally, a 27-year-old with three years of education reported a salary of $22,000."

Examples: text documents, images, videos, etc.

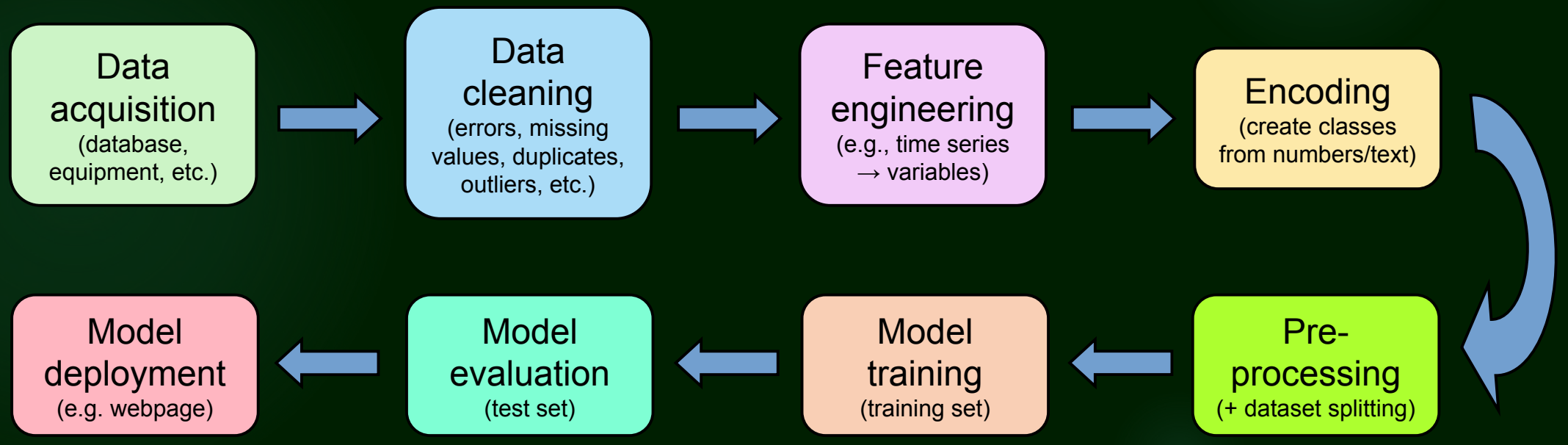
Data

Data quality

1) The ML model is just as good as the data used to train it!

2) There is no guarantee to obtain a good ML model, even for high-quality data!

Data (+ML) workflow (simplified!)



Data

General terms

Dataset

5 Samples

Age	Education time	Salary	Permanent?
20	2	17000	No
45	10	37000	Yes
16	0	12000	No
34	5	26000	Yes
27	3	22000	No

2 Features

2 Targets

„Permanent“ is a categorical column, the others are numeric

Dataset splitting

Training set (e.g., 70-80% of the dataset)

Test set (e.g., 20-30% of the dataset)

Train the ML model with the **training set**, evaluate (or test) the model with the **test set**

Ethics

→ **Challenging!** Under discussion/development!

Bias and Fairness

- Algorithms can perpetuate or amplify existing social biases.

Solution: Use a representative, balanced and fair dataset to train your model!

Transparency and explainability

- How ML systems make decisions?
- Trade-off model complexity <=> interpretability.

Solution: Use the simplest model that solves the problem. Try to explain it (and state its limitations!) when showing it to others.

Environmental impact

- Energy consumption of training large models (e.g.: save trained models & large arrays for later use)
- Carbon footprint considerations in ML design (e.g.: reuse the code, share the ML model and dataset)

Ethics

Privacy and data rights

- Data collection needs versus individual privacy
- Data anonymization as a requirement

Accountability and responsibility

- Who is responsible when an ML system causes harm?
- Discussion of governance frameworks and oversight

Solutions:

- Impact assessments (by ML committees) before deploying high-risk ML systems
- Involvement with or development of broad ML regulations like *EU AI Act*

Real example: FDA Regulation of ML in Healthcare: The FDA agency (USA) has developed frameworks for evaluating and approving ML-based medical devices, balancing innovation with patient safety.

Case studies

Optimize polymer compositions to get high target properties

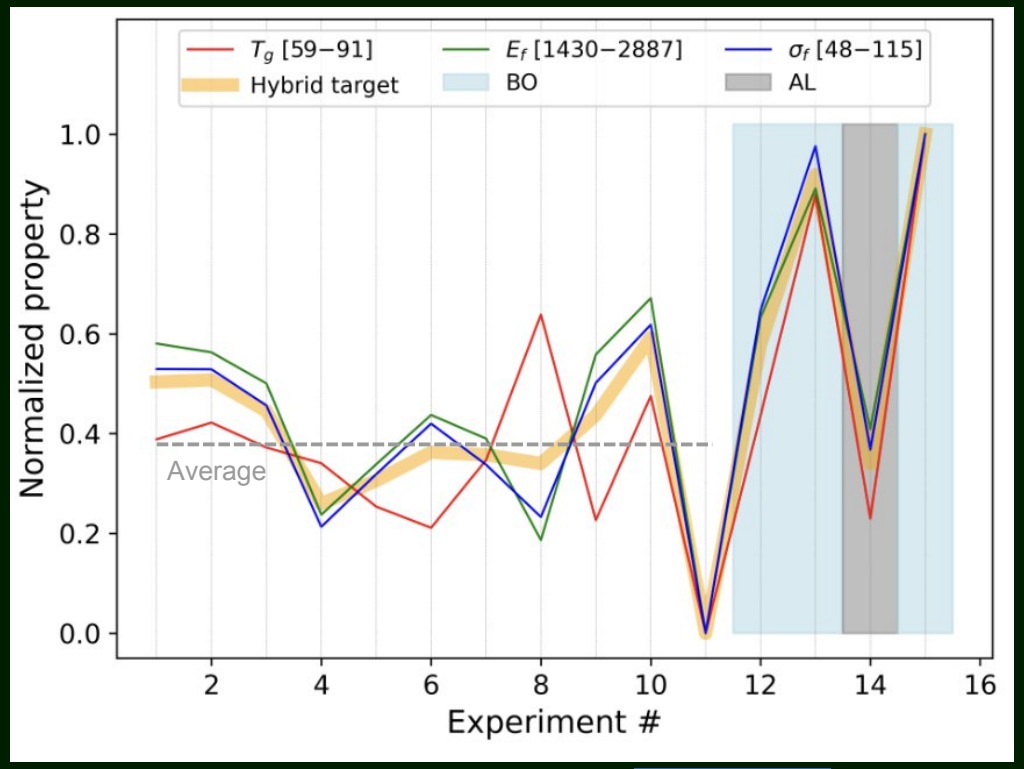
	Name	BC
Resins	DGEBF	0
	NC-514S	68
	PP1314	99
	PP1402	100
Hardeners	IPDA	0
	DETDA	0
	LITE2002	66
	LITE2402	34.3

Maximize:

- Glass transition temperature (T_g)
- Flexural modulus (E_f)
- Flexural strength (σ_f)

Restrictions: *20 % ≤ Biocontent (BC) ≤ 30 %*

ML



Best experiment (# 15)

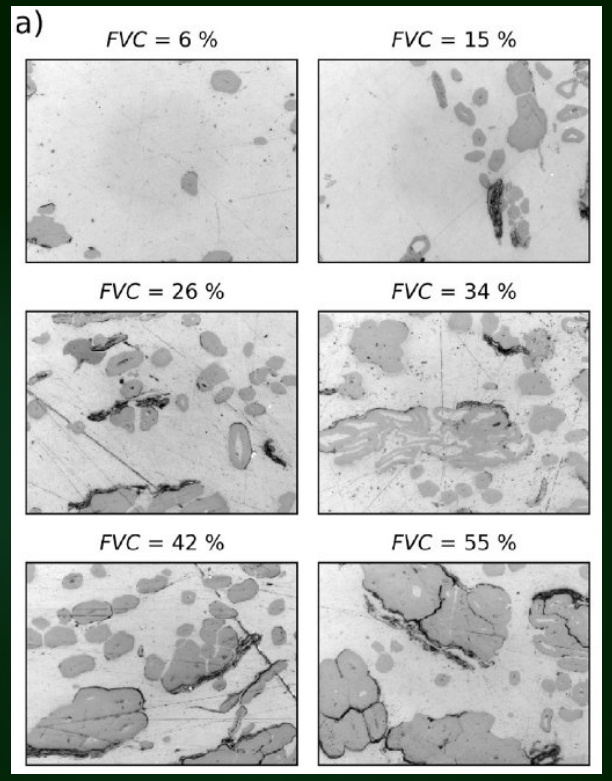
Biocontent = 20 %

67 %	DGEBF
1 %	NC-514S
0 %	PP1314
4 %	PP1402
0 %	IPDA
0 %	DETDA
18 %	LITE2002
10 %	LITE2402

Albuquerque *et al.*, *ACS Appl. Eng. Mater.* 1 (**2023**) 3298−3308.

Case studies

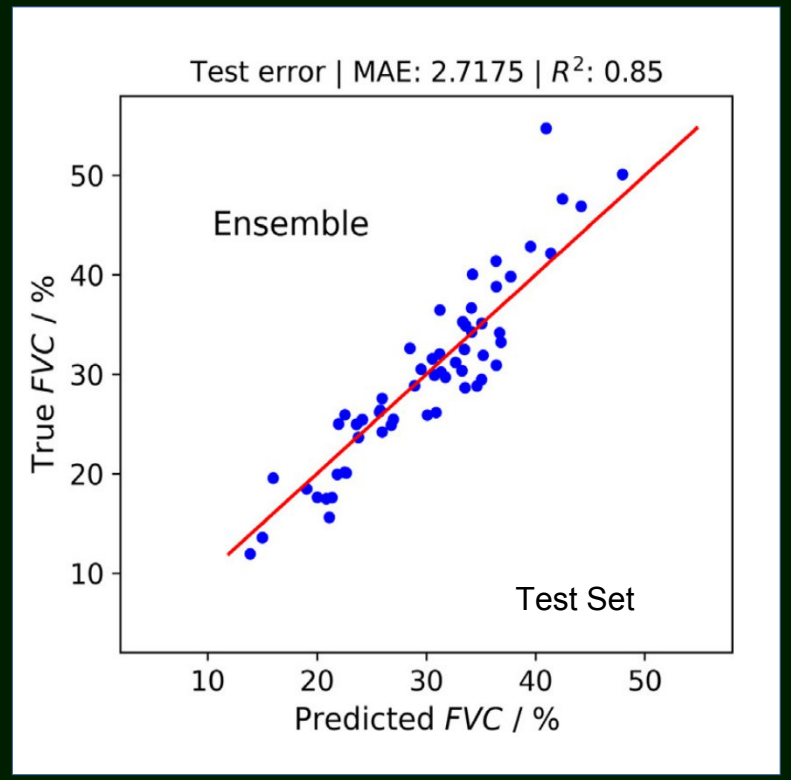
Characterization of composites



290 SEM images

CNN training , XGBoost training

Parity plot

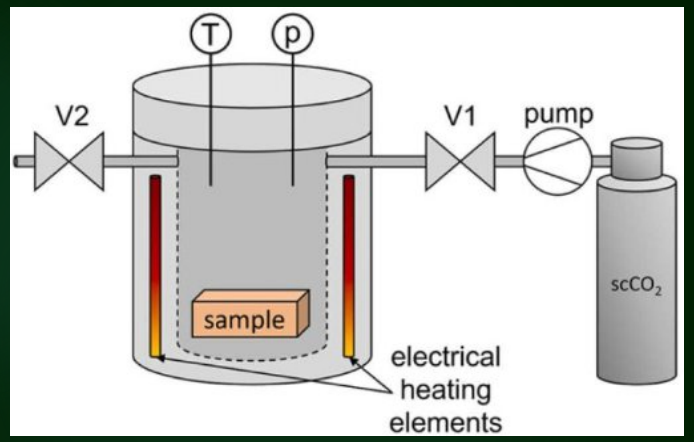


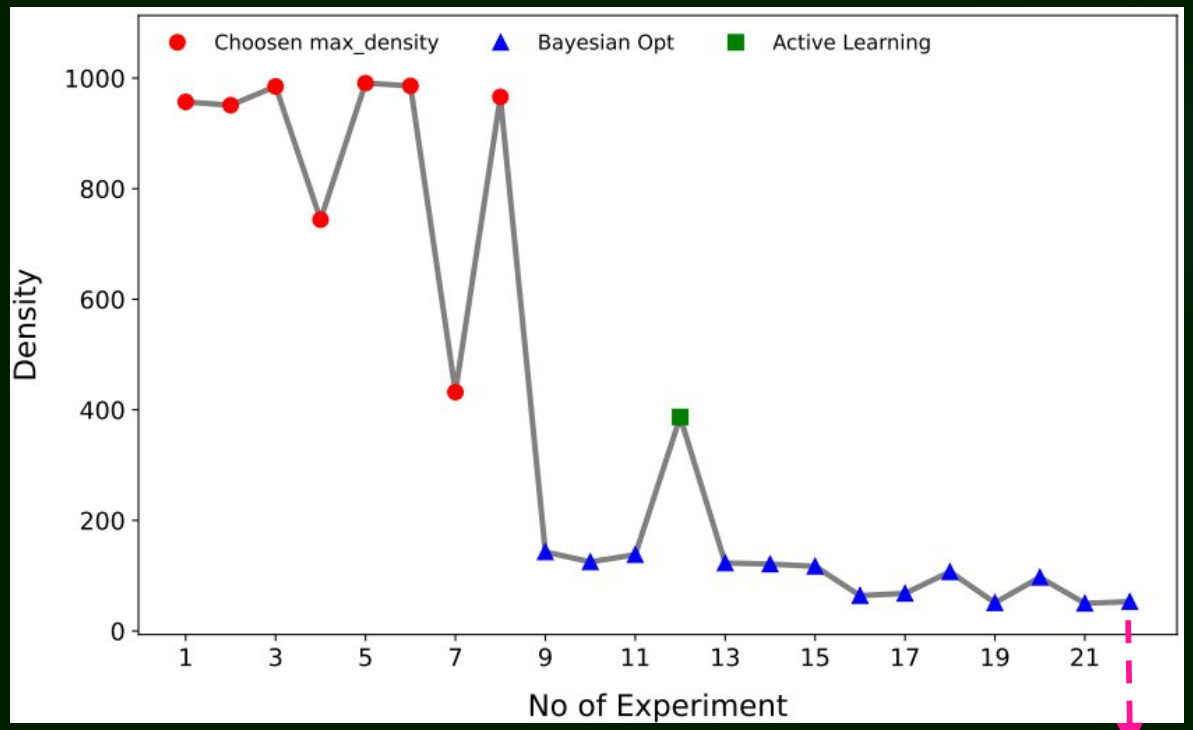
Rothenhäusler et al., *Machine Learning with Applications* 19 (**2025**) 100609.

Case studies

Minimize the density of a polymer foam

- PA12 foam
- Params: **p, T, time**





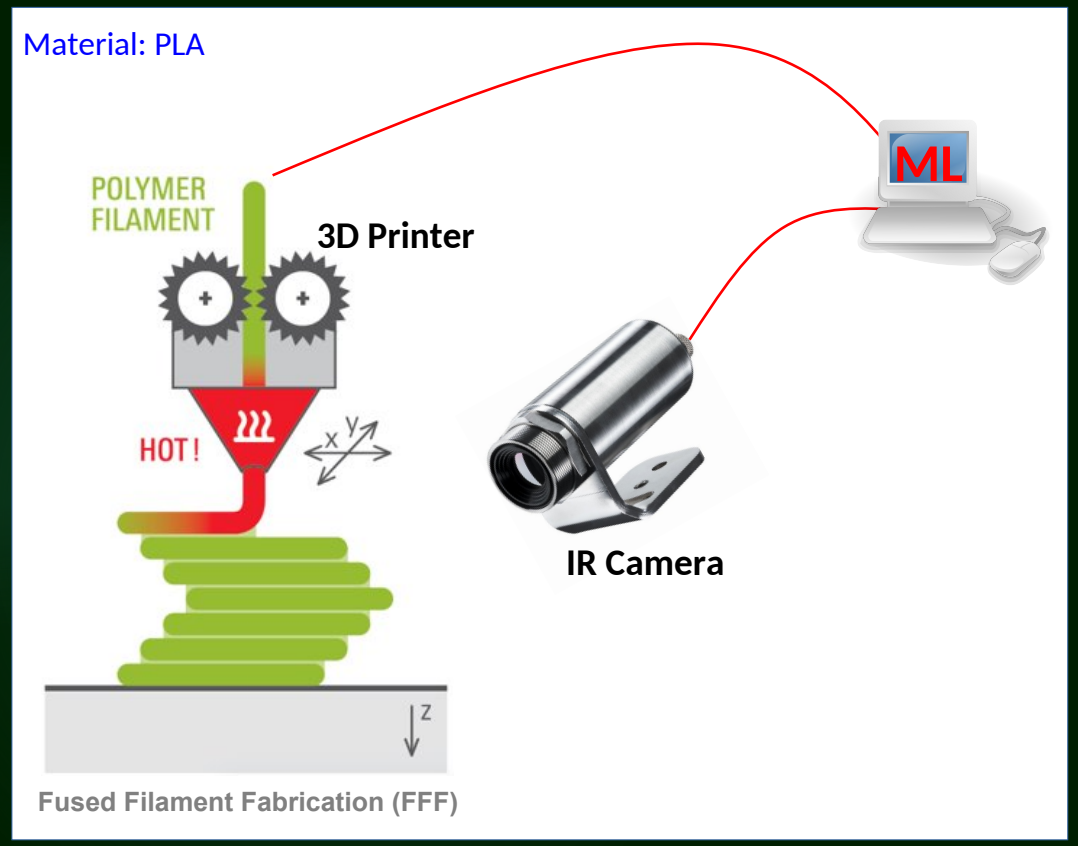
(p = 170 bar, T = 188° C, time = 20 min)

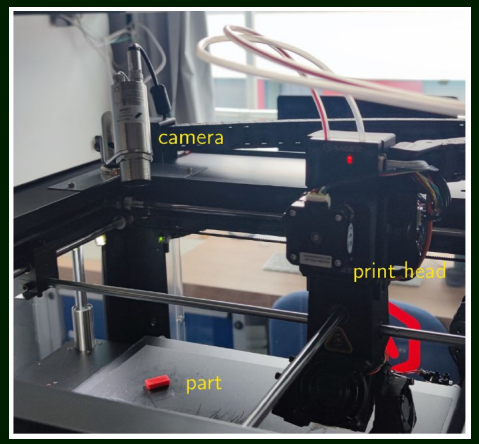
50 kg/m^3

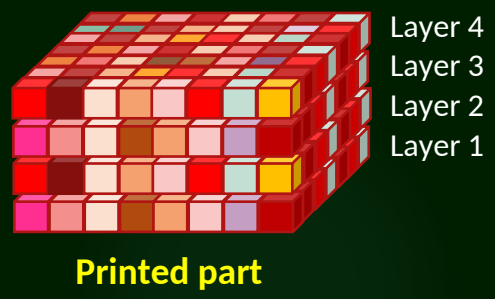
Shah *et al.*, *Polymer* 320 (**2025**) 128096.

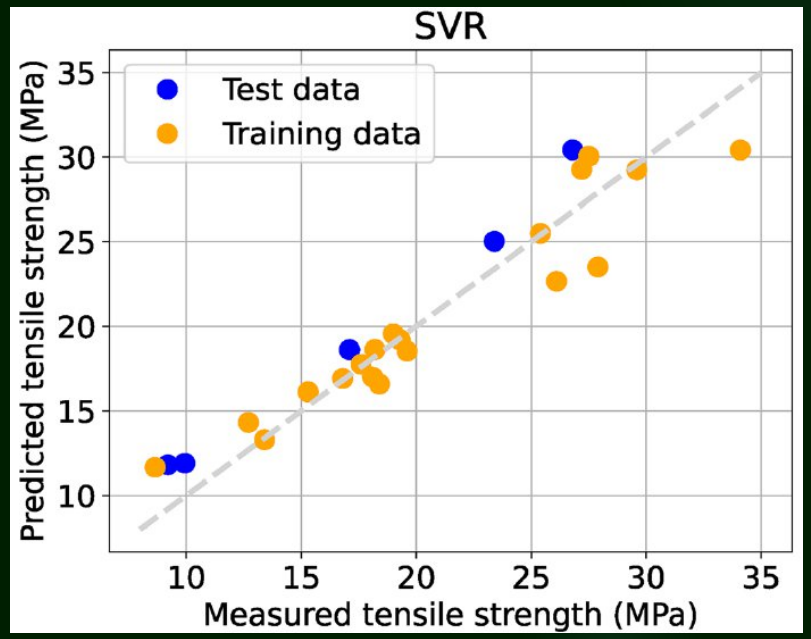
Case studies

Predict future mechanical properties from IR images









N. Bauriedel, et al., *J. Pol. Sci.* 62 (**2024**) 5633-5641.

Exercises

1) Open a gmail account (**or** be sure to have jupyter notebook installed locally)
2) Download the file ‘files1.zip’ and unzip it locally on your computer
3) Go to google colab (online) **or** launch jupyter notebook locally (offline)
4) Open/upload ‘exercise1.ipynb’ and wait for more instructions